

Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

Branches of $\log f$ on simply connected domains — with spaced layout

Problem

Let U be a simply connected domain in the complex plane and let f be holomorphic on U with $f(z) \neq 0$ for all $z \in U$.

- (1) Show that there exists a holomorphic F on U with $e^F = f$ (a branch of $\log f$ on U).
 - (2) Give an example showing that a holomorphic branch of the *scalar* logarithm need not exist on the image $f(U)$.
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Background and tools

(i) Logarithms via primitives.

If f is holomorphic and never zero on a domain D , then $\frac{f'}{f}$ is holomorphic on D . If there exists a holomorphic F with $F' = \frac{f'}{f}$, then

$$\frac{(e^F)'}{e^F} = F' = \frac{f'}{f} \quad \implies \quad \left(\log \frac{e^F}{f} \right)' = 0,$$

so e^F/f is constant; adding a constant to F we obtain $e^F = f$.

(ii) Primitives on simply connected domains.

If g is holomorphic on a simply connected domain U , then g has a global primitive G on U ; equivalently,

$$\oint_{\gamma} g(z) dz = 0 \quad \text{for every closed loop } \gamma \subset U.$$

(iii) Integral and winding criteria.

For any closed loop γ in $\{f \neq 0\}$,

$$\oint_{\gamma} \frac{f'(z)}{f(z)} dz = 2\pi i \cdot \text{Wind}(f \circ \gamma, 0).$$

Thus a holomorphic branch of $\log f$ on a domain $D \subset \{f \neq 0\}$ exists iff the integral above is 0 for all closed loops in D (equivalently, $f \circ \gamma$ never winds around 0).

(iv) Scalar log on a planar domain.

A holomorphic branch of the scalar logarithm on a domain $V \subset \mathbf{C} \setminus \{0\}$ exists iff V is simply connected (i.e. V does not wrap around the origin). Annuli about 0 and $\mathbf{C} \setminus \{0\}$ itself fail this.

Solution to (1): Existence on U

Because f is holomorphic and nonvanishing, the function

$$g(z) := \frac{f'(z)}{f(z)}$$

is holomorphic on U . Since U is simply connected, there exists a holomorphic primitive G with $G' = g$ on U .

Define

$$H(z) := e^{G(z)}.$$

Then

$$\frac{H'(z)}{H(z)} = G'(z) = \frac{f'(z)}{f(z)} \quad \implies \quad \left(\log \frac{H(z)}{f(z)} \right)' = 0.$$

Hence H/f is constant on U . Choose $c \in \mathbf{C}$ with $e^c = f(z_0)/H(z_0)$ for some (hence every) $z_0 \in U$, and set

$$F(z) := G(z) + c.$$

Then $e^{F(z)} = f(z)$ on U , so F is a holomorphic branch of $\log f$.

Uniqueness up to $2\pi i\mathbf{Z}$. If F_1 and F_2 both satisfy $e^F = f$, then $e^{F_1 - F_2} \equiv 1$, so $F_1 - F_2$ is a constant of the form $2\pi i k$ with $k \in \mathbf{Z}$.

Equivalent criteria on U

Integral criterion. A holomorphic F with $e^F = f$ exists on U if and only if

$$\oint_{\gamma} \frac{f'(z)}{f(z)} dz = 0 \quad \text{for every closed loop } \gamma \subset U.$$

Winding criterion. Equivalently, a holomorphic branch of $\log f$ exists on U iff

$$\text{Wind}(f \circ \gamma, 0) = 0 \quad \text{for every closed loop } \gamma \subset U.$$

Solution to (2): Examples where a scalar log on $f(U)$ fails

Global example. Let $U = \mathbf{C}$ and $f(z) = e^z$. Then f is entire, never zero, and a branch $\log f$ exists on U (take $F(z) = z$). However

$$f(U) = \mathbf{C} \setminus \{0\},$$

which is not simply connected. For the unit circle $\Gamma = \{|w| = 1\}$,

$$\oint_{\Gamma} \frac{dw}{w} = 2\pi i \neq 0.$$

If a holomorphic scalar logarithm existed on $\mathbf{C} \setminus \{0\}$, its derivative would be $1/w$, forcing this integral to be 0 for all closed loops—a contradiction.

Bounded variant (annulus image). Let $U = \{z : |z| < 2\}$ and $f(z) = e^z$. Then $\log f$ exists on U (again $F(z) = z$), but

$$f(U) = \{e^{x+iy} : |x| < 2, |y| < 2\} = \{w : e^{-2} < |w| < e^2\},$$

which is an annulus encircling 0. Any circle $\{|w| = r\}$ with $e^{-2} < r < e^2$ satisfies

$$\oint \frac{dw}{w} = 2\pi i \neq 0,$$

so no holomorphic scalar logarithm exists on $f(U)$.

Geometric/topological intuition

The exponential map $\exp : \mathbf{C} \rightarrow \mathbf{C} \setminus \{0\}$ is a covering map with deck transformations $z \mapsto z + 2\pi i k$ ($k \in \mathbf{Z}$). A global holomorphic scalar logarithm on a target domain $V \subset \mathbf{C} \setminus \{0\}$ would be a single-valued holomorphic inverse on V ; this exists exactly when every loop in V lifts to a closed loop, i.e. when V is simply connected. Annuli and $\mathbf{C} \setminus \{0\}$ are not, so monodromy around 0 prevents a single-valued holomorphic logarithm there.

For $\log f$ on U , the only obstruction would be that some image loop $f(\gamma)$ winds around 0. On simply connected U with $f \neq 0$, the form f'/f is exact, so all such windings vanish and a branch exists.

Summary

On any simply connected domain U , every holomorphic nonvanishing f admits a holomorphic F with $e^F = f$ (unique up to $2\pi i \mathbf{Z}$). Nevertheless, a holomorphic scalar logarithm on $f(U)$ can fail to exist when $f(U)$ winds around the origin (e.g. $f = \exp$ gives $f(U) = \mathbb{C} \setminus \{0\}$ or an annulus).

Example. Let $U = \mathbb{C}$ and $f(z) = e^z$.

Existence of $\log f$ on U . Since f is entire and never zero, $f'/f \equiv 1$ is entire. Because U is simply connected, $G(z) := z$ is a primitive of f'/f . Define $F(z) := G(z) = z$. Then $e^{F(z)} = e^z = f(z)$, so F is a holomorphic branch of $\log f$ on U . Uniqueness holds up to $2\pi i \mathbf{Z}$: if F_1, F_2 satisfy $e^{F_j} = f$, then $e^{F_1 - F_2} \equiv 1$, hence $F_1 - F_2 \in 2\pi i \mathbf{Z}$ is a constant.

Failure of a branch of $\log$ on $f(U)$. We have $f(U) = \mathbb{C}^\times$. If there were a holomorphic branch L of the scalar logarithm on $\mathbb{C}^\times$, then $L'(w) = 1/w$ and, for every closed loop $\Gamma \subset \mathbb{C}^\times$,

$$\oint_{\Gamma} \frac{dw}{w} = \oint_{\Gamma} L'(w) dw = 0.$$

Taking $\Gamma = \{|w| = 1\}$ yields $\oint_{|w|=1} \frac{dw}{w} = 2\pi i \neq 0$, a contradiction. Therefore, no holomorphic branch of the scalar log exists on $\mathbb{C}^\times$.

Monodromy/winding viewpoint. For any closed loop Γ in a domain that avoids 0,

$$\oint_{\Gamma} \frac{dw}{w} = 2\pi i \cdot \text{Wind}(\Gamma, 0).$$

Thus a global holomorphic branch of $\log$ on a domain $V \subset \mathbb{C}^\times$ exists iff every loop in V has winding number 0 about 0 (equivalently, V is simply connected). Here $V = \mathbb{C}^\times$ has nontrivial loops around 0, so monodromy obstructs single-valuedness.

Analytic continuation around a circle (explicit jump). Start with a local branch of $\log$ near $w = 1$ and analytically continue once along the unit circle $w = e^{it}$, $t \in [0, 2\pi]$. The continuation returns to the starting point but accumulates the increment

$$\int_0^{2\pi} \frac{d}{dt}(\log(e^{it})) dt = \int_0^{2\pi} i dt = 2\pi i,$$

so the value jumps by $2\pi i$. This nontrivial monodromy forbids a single-valued holomorphic logarithm on $\mathbb{C}^\times$.

Bounded variant (annulus image). Let $U = \{z : |z| < 2\}$ and $f(z) = e^z$. Then $F(z) = z$ gives a branch of $\log f$ on U , but

$$f(U) = \{e^{x+iy} : |x| < 2, |y| < 2\} = \{w \in \mathbb{C}^\times : e^{-2} < |w| < e^2\},$$

an annulus encircling 0. Any circle $\{|w| = r\}$ with $e^{-2} < r < e^2$ satisfies $\oint \frac{dw}{w} = 2\pi i \neq 0$, so no holomorphic branch of the scalar log exists on $f(U)$.

Local vs. global behavior. On any simply connected subdomain $V \subset \mathbb{C}^\times$ that does not wrap around 0 (e.g. a slit plane or a sector of angle $< 2\pi$), one *can* choose a holomorphic branch of the scalar log. The obstruction is purely topological: encircling 0 forces a $2\pi i$ jump.

Connection with $\arg$ and f'/f . For $w \neq 0$, $\log w = \log |w| + i \arg w$; the impossibility of a global continuous $\arg$ on $\mathbb{C}^\times$ is the same obstruction. Equivalently, if L existed on V , then $(\log \circ f)' = (f'/f)$ would be the derivative of a *global* holomorphic function on V . On simply connected U this always holds for f'/f (hence $\log f$ exists on U); on $f(U)$ it fails exactly when some loop winds around 0.

Takeaway. A holomorphic nonvanishing f on a simply connected U always has a holomorphic logarithm F with $e^F = f$ (unique up to $2\pi i\mathbb{Z}$). But a holomorphic scalar logarithm on $f(U)$ may not exist if $f(U)$ winds around 0 (as with $f = \exp$ giving $\mathbb{C}^\times$ or an annulus).

- f is entire and never zero, so $f'/f \equiv 1$ is entire.
- On the simply connected set $U = \mathbb{C}$, a primitive of f'/f is $G(z) = z$.
- Set $F(z) := G(z) = z$. Then $e^{F(z)} = e^z = f(z)$. Thus F is a (holomorphic) branch of $\log f$ on U .

2) Why a holomorphic log cannot exist on $f(U) = \mathbb{C}^\times$

- $f(U) = \mathbb{C} \setminus \{0\}$ is not simply connected (its fundamental group is $\mathbb{Z}$); any loop going once around 0 has nonzero winding number.
- If a holomorphic branch of the scalar logarithm Log existed on $\mathbb{C}^\times$, then $(\text{Log})'(w) = 1/w$ and hence, for every closed loop $\Gamma \subset \mathbb{C}^\times$,

$$\oint_{\Gamma} \frac{dw}{w} = \oint_{\Gamma} (\text{Log})'(w) dw = 0.$$

But for the unit circle $\{|w| = 1\}$,

$$\oint_{|w|=1} \frac{dw}{w} = 2\pi i \neq 0.$$

Contradiction. Therefore no holomorphic branch of $\log$ exists on $\mathbb{C}^\times$.

3) Geometry/topology intuition (why the obstruction appears)

- $\exp : \mathbb{C} \rightarrow \mathbb{C}^\times$ is a *universal covering map*. Adding $2\pi i$ in z -space leaves e^z unchanged: $e^{z+2\pi i} = e^z$. So loops around 0 in $\mathbb{C}^\times$ lift to paths in $\mathbb{C}$ whose endpoints differ by $2\pi i$.

- A single-valued holomorphic log on $\mathbb{C}^\times$ would be a global holomorphic inverse to $\exp$; that would force all lifted loops to close, which they do not—this is the *monodromy* obstruction.
- Equivalent “metric picture”: horizontal lines $y = c$ map to rays of angle c ; vertical lines $x = c$ map to circles $|w| = e^c$. The image wraps endlessly around 0; choosing a continuous argument globally is impossible.

Variant (same phenomenon). Take $U = \{|z| < 2\}$ and $f(z) = e^z$. Then

$$f(U) = \{e^{x+iy} : |x| < 2, |y| < 2\} = \{w \in \mathbb{C}^\times : e^{-2} < |w| < e^2\},$$

an annulus around 0. The same integral test around any circle in that annulus, $\oint \frac{dw}{w} = 2\pi i$, blocks a global holomorphic log there.

Cauchy integral with principal value and boundary limits

Problem (restated)

Let Γ be a simple smooth positively oriented closed curve in the plane, and let f be continuously differentiable on Γ . Define, for $z \in \mathbb{C} \setminus \Gamma$,

$$F(z) := \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

This integral diverges when z approaches a point $\zeta_0 \in \Gamma$ with $f(\zeta_0) \neq 0$. Define the principal value at ζ_0 by

$$\text{P.V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta := \lim_{\varepsilon \rightarrow 0^+} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta,$$

where Γ_ε is Γ with a small open arc of radius ε around ζ_0 removed (the remaining part is traversed with the original orientation).

Background and basic facts

Analyticity off the curve. For $z \notin \Gamma$ the integrand is continuous on Γ and the function F is well defined and holomorphic in each of the two components of $\mathbb{C} \setminus \Gamma$ (the interior and the exterior). Differentiation under the integral sign gives

$$F'(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta, \quad z \in \mathbb{C} \setminus \Gamma.$$

Recovery of interior analytic data. If f extends to a function holomorphic in the interior of Γ and continuous on $\overline{\text{int}(\Gamma)}$, then the Cauchy integral formula yields

$$F(z) = f(z) \quad \text{for } z \text{ in the interior of } \Gamma, \quad F(z) = 0 \quad \text{for } z \text{ in the exterior and } |z| \text{ large.}$$

Principal value meaning. At $z = \zeta_0 \in \Gamma$ the kernel $(\zeta - \zeta_0)^{-1}$ has a simple pole on the contour. The principal value extracts the finite, symmetric part of the singular integral by removing a small arc around the pole and letting its size shrink to 0.

Nontangential limits and the Sokhotski–Plemelj formulas

Let $F^\pm(\zeta_0)$ denote the limits of $F(z)$ as $z \rightarrow \zeta_0 \in \Gamma$ from the interior (+) and from the exterior (−) along nontangential approaches. If $f \in C^1(\Gamma)$, then these limits exist and

$$\begin{aligned} F^+(\zeta_0) &= \frac{1}{2} f(\zeta_0) + \text{P.V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta, \\ F^-(\zeta_0) &= -\frac{1}{2} f(\zeta_0) + \text{P.V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta. \end{aligned}$$

Consequently,

$$\boxed{F^+(\zeta_0) - F^-(\zeta_0) = f(\zeta_0)} \quad \text{and} \quad \boxed{\frac{F^+(\zeta_0) + F^-(\zeta_0)}{2} = \text{P.V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta.}$$

Sketch of derivation (indentation method)

Fix $\zeta_0 \in \Gamma$ and replace a small arc of Γ near ζ_0 by a circle of radius ε centered at ζ_0 , lying inside (for F^+) or outside (for F^-) the domain. Then

$$F(z) = \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta + \frac{1}{2\pi i} \int_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta,$$

where C_ε is the small circular arc. Let $z \rightarrow \zeta_0$ from the chosen side. The first integral tends to the principal value term. On C_ε , write $f(\zeta) = f(\zeta_0) + o(1)$ and compute

$$\frac{1}{2\pi i} \int_{C_\varepsilon} \frac{f(\zeta_0)}{\zeta - \zeta_0} d\zeta = \pm \frac{1}{2} f(\zeta_0),$$

with the sign + for the inner indentation and − for the outer one (by orientation). Adding the contributions yields the two Plemelj formulas above.

Remarks

- The regularity assumption $f \in C^1(\Gamma)$ can be weakened (e.g. Hölder continuity suffices for the nontangential limits).
- If Γ is a line, these formulas reduce to the classical Hilbert transform jump relations.
- Orientation matters: reversing the orientation of Γ switches the signs in the $\pm \frac{1}{2} f(\zeta_0)$ terms.

Concrete example on the unit circle. Let $\Gamma = \{\zeta : |\zeta| = 1\}$ (counterclockwise) and $f(\zeta) = \zeta^m$ with $m \in \mathbb{Z}_{\geq 0}$. Define

$$F(z) = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z} d\zeta, \quad z \in \mathbb{C} \setminus \Gamma.$$

Inside/outside values. By Cauchy's integral formula, if $|z| < 1$ then $F(z) = z^m$, while if $|z| > 1$ then $F(z) = 0$. Thus F is analytic on each side of Γ , equals z^m in the interior, and vanishes in the exterior.

Boundary limits and principal value. For $\zeta_0 \in \Gamma$, the Sokhotski–Plemelj formulas yield

$$F^+(\zeta_0) = \frac{1}{2}\zeta_0^m + \text{P.V.} \cdot \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0} d\zeta, \quad F^-(\zeta_0) = -\frac{1}{2}\zeta_0^m + \text{P.V.} \cdot \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0} d\zeta.$$

Since $F^+(\zeta_0) = \zeta_0^m$ and $F^-(\zeta_0) = 0$, it follows that

$$\boxed{\text{P.V.} \cdot \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0} d\zeta = \frac{1}{2}\zeta_0^m}, \quad \boxed{F^+(\zeta_0) - F^-(\zeta_0) = \zeta_0^m = f(\zeta_0)}.$$

Special case $m = 0$. For $f(\zeta) \equiv 1$ we have $F(z) = 1$ for $|z| < 1$ and $F(z) = 0$ for $|z| > 1$, and on Γ ,

$$F^+(\zeta_0) = 1, \quad F^-(\zeta_0) = 0, \quad \text{P.V.} \cdot \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta - \zeta_0} d\zeta = \frac{1}{2}.$$

Inside/outside values (unit circle $\Gamma = \{|\zeta| = 1\}$). Consider

$$F(z) = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z} d\zeta, \quad m \in \mathbb{Z}_{\geq 0}.$$

Case $|z| < 1$ (pole inside). As a function of ζ , the integrand has a simple pole at $\zeta = z$ lying inside Γ . By Cauchy's integral formula,

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z} d\zeta = \zeta^m|_{\zeta=z} = z^m.$$

Equivalently, $\text{Res}_{\zeta=z} \frac{\zeta^m}{\zeta - z} = z^m$, so the integral is $2\pi i z^m$, and dividing by $2\pi i$ yields $F(z) = z^m$.

Case $|z| > 1$ (no pole inside). Now the pole $\zeta = z$ lies outside Γ , hence the integrand is holomorphic on and inside the contour. By Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z} d\zeta = 0, \quad \text{i.e. } F(z) = 0.$$

Series check for $|z| > 1$ (optional). For $|z| > 1$,

$$\frac{\zeta^m}{\zeta - z} = -\frac{1}{z} \frac{\zeta^m}{1 - \zeta/z} = -\frac{1}{z} \sum_{k=0}^{\infty} \frac{\zeta^{m+k}}{z^k} \quad (|\zeta/z| < 1).$$

Integrating termwise over $|\zeta| = 1$ gives $\oint \zeta^n d\zeta = 0$ for all $n \neq -1$, so the whole integral is 0, confirming $F(z) = 0$ outside.

Cauchy principal value on a smooth curve: existence and formula

Problem (exact restatement of the two tasks)

(a) Show that the principal value is well defined for all $\zeta_0 \in \Gamma^*$, namely that the limit

$$\text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta := \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta$$

exists, where Γ is a simple smooth positively oriented curve, f is continuously differentiable on Γ^* , and Γ_ε is the part of Γ that is at least ε -away from ζ_0 .

(b) Show that

$$\text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{1}{2} f(\zeta_0).$$

Background and notation

Let Γ be a simple C^1 positively oriented closed curve in the plane and let $f \in C^1(\Gamma)$. For $z \in \mathbb{C} \setminus \Gamma$ define the Cauchy transform

$$F(z) := \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Then F is holomorphic on each component of $\mathbb{C} \setminus \Gamma$. As $z \rightarrow \zeta_0 \in \Gamma$, the kernel $(\zeta - z)^{-1}$ has a simple pole on the contour; the principal value removes a small arc around ζ_0 and takes the limit as its length shrinks to 0.

Solution to (a): existence of the principal value

Fix $\zeta_0 \in \Gamma$. Split the integrand into a regular part and a pure singular kernel:

$$\frac{f(\zeta)}{\zeta - \zeta_0} = \underbrace{\frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}}_{\text{regular near } \zeta_0} + \underbrace{\frac{f(\zeta_0)}{\zeta - \zeta_0}}_{\text{pure kernel}}. \quad (1)$$

Regular part. Since $f \in C^1(\Gamma)$, there is a constant C with

$$|f(\zeta) - f(\zeta_0)| \leq C |\zeta - \zeta_0| \quad \text{for } \zeta \text{ near } \zeta_0.$$

Hence the quotient $\frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}$ is bounded and continuous on Γ (interpreted by continuity at ζ_0). Therefore

$$\int_{\Gamma_\varepsilon} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta \quad \text{converges as } \varepsilon \downarrow 0,$$

so its contribution to the principal value has a proper limit (indeed, it is just the ordinary integral over Γ).

Pure kernel. On the small missing arc of radius ε around ζ_0 the integral

$$\int \frac{d\zeta}{\zeta - \zeta_0}$$

is $\pm\pi i$ in the limit (half of the $2\pi i$ residue), with the sign determined by whether the indentation is taken inside or outside, together with the positive orientation of Γ . Thus the contribution of the pure kernel to the truncated integral is uniformly bounded and tends, in the principal value sense, to the finite number $\frac{1}{2} 2\pi i \cdot \frac{f(\zeta_0)}{2\pi i} = \frac{1}{2} f(\zeta_0)$ (precisely accounted for in part (b) below).

Combining the two parts from (1) shows that the limit defining

$$\text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta$$

exists for every $\zeta_0 \in \Gamma$.

Solution to (b): decomposition formula with the $\frac{1}{2}f(\zeta_0)$ term

Start from the splitting (1) and integrate over Γ_ε :

$$\frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{f(\zeta_0)}{2\pi i} \int_{\Gamma_\varepsilon} \frac{d\zeta}{\zeta - \zeta_0}.$$

Let $\varepsilon \downarrow 0$. The first term converges to the *ordinary* integral over Γ because its integrand is bounded and continuous on Γ :

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta.$$

For the second term,

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{d\zeta}{\zeta - \zeta_0} = \frac{1}{2},$$

because the missing small arc carries half of the $2\pi i$ residue at $\zeta = \zeta_0$ (with the positive orientation prescribed in the statement). Therefore,

$$\text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{1}{2} f(\zeta_0),$$

as required.

Concrete example on the unit circle

Let $\Gamma = \{\zeta : |\zeta| = 1\}$ (counterclockwise), $\zeta_0 = e^{it_0}$, and $f(\zeta) = \zeta^m$ with $m \geq 0$. Then

$$\text{P.V.} \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m - \zeta_0^m}{\zeta - \zeta_0} d\zeta + \frac{1}{2} \zeta_0^m.$$

But the integrand on the right is regular on the unit circle, and Cauchy's formula gives the value

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m - \zeta_0^m}{\zeta - \zeta_0} d\zeta = \frac{1}{2} \zeta_0^m,$$

so the principal value equals $\frac{1}{2} \zeta_0^m$. This matches the Plemelj jump relations: the interior limit is ζ_0^m and the exterior limit is 0, hence their average is exactly the principal value.

Connection between the principal value and the index of the curve

For a smooth closed curve Γ and a point ζ_0 not on Γ , the *index* (or winding number) of Γ about ζ_0 is defined by

$$\text{Ind}(\Gamma; \zeta_0) := \frac{1}{2\pi i} \oint_{\Gamma} \frac{d\zeta}{\zeta - \zeta_0}.$$

It counts how many times Γ winds around ζ_0 , with orientation. For a positively oriented simple Jordan curve,

$$\text{Ind}(\Gamma; \zeta_0) = \begin{cases} 1, & \zeta_0 \text{ inside } \Gamma, \\ 0, & \zeta_0 \text{ outside } \Gamma. \end{cases}$$

When $\zeta_0 \in \Gamma$, the integral above diverges because of the singularity at $\zeta = \zeta_0$. However, its *Cauchy principal value* exists and equals *half* the index:

$$\text{P.V.} \frac{1}{2\pi i} \oint_{\Gamma} \frac{d\zeta}{\zeta - \zeta_0} = \frac{1}{2} \text{Ind}(\Gamma; \zeta_0).$$

For a positively oriented simple curve, $\text{Ind}(\Gamma; \zeta_0) = 1$, so

$$\text{P.V.} \frac{1}{2\pi i} \oint_{\Gamma} \frac{d\zeta}{\zeta - \zeta_0} = \frac{1}{2}.$$

General decomposition in terms of the index

In the decomposition formula

$$\text{P.V.} \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{1}{2} f(\zeta_0),$$

the last term $\frac{1}{2} f(\zeta_0)$ comes precisely from the principal value of the pure kernel. Replacing the constant $\frac{1}{2}$ by $\frac{1}{2} \text{Ind}(\Gamma; \zeta_0)$ gives the more general form valid for any winding number:

$$\boxed{\text{P.V.} \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} d\zeta + \frac{\text{Ind}(\Gamma; \zeta_0)}{2} f(\zeta_0).}$$

Consequences for boundary limits (Plemelj jump)

Define

$$F(z) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad z \in \mathbb{C} \setminus \Gamma.$$

Then the nontangential limits from the interior (+) and exterior (−) satisfy

$$F^{\pm}(\zeta_0) = \text{P. V.} \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta \pm \frac{\text{Ind}(\Gamma; \zeta_0)}{2} f(\zeta_0), \quad \zeta_0 \in \Gamma.$$

Hence the *jump relation*

$$\boxed{F^+(\zeta_0) - F^-(\zeta_0) = \text{Ind}(\Gamma; \zeta_0) f(\zeta_0).}$$

Remarks

- For a positively oriented simple contour, $\text{Ind}(\Gamma; \zeta_0) = 1$, so the jump equals $f(\zeta_0)$ and the principal value term corresponds to $\frac{1}{2}f(\zeta_0)$.
- If Γ winds k times around ζ_0 , then $\text{Ind}(\Gamma; \zeta_0) = k$, and all the $\frac{1}{2}$ coefficients are multiplied by k .
- When ζ_0 lies outside Γ , $\text{Ind} = 0$, so the principal value is simply $\frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta = 0$.

(c) Interior/Exterior limits and the jump

Let $F(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta$ with Γ a simple C^1 Jordan curve and $f \in C^1(\Gamma)$. For $\zeta_0 \in \Gamma$,

$$\lim_{\substack{z \rightarrow \zeta_0 \\ z \text{ inside } \Gamma}} F(z) = \text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta + \frac{1}{2}f(\zeta_0),$$

$$\lim_{\substack{z \rightarrow \zeta_0 \\ z \text{ outside } \Gamma}} F(z) = \text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta - \frac{1}{2}f(\zeta_0).$$

Thus F has a jump $F^+(\zeta_0) - F^-(\zeta_0) = f(\zeta_0)$.

Proof sketch. Split $\frac{f(\zeta)}{\zeta - \zeta_0} = \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} + \frac{f(\zeta_0)}{\zeta - \zeta_0}$. The first term yields the principal value. The second contributes $\pm \frac{1}{2}f(\zeta_0)$ via an indentation around ζ_0 (the short circular arc integral equals $\pm \pi i$).

(d) What if Γ is not simple / only piecewise smooth? What if f is not C^1 ?

Non-simple curves (multiple windings). Replace $\pm \frac{1}{2}$ by $\pm \frac{1}{2} \text{Ind}(\Gamma; \zeta_0)$:

$$F^{\pm}(\zeta_0) = \text{P. V.} \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta \pm \frac{\text{Ind}(\Gamma; \zeta_0)}{2} f(\zeta_0), \quad F^+ - F^- = \text{Ind}(\Gamma; \zeta_0) f(\zeta_0).$$

Piecewise smooth / Lipschitz curves. For Γ Lipschitz (e.g. piecewise C^1), the same formulas hold for *nontangential* limits. The Cauchy singular integral operator is bounded on $L^p(\Gamma)$, $1 < p < \infty$, so Plemelj limits exist a.e.

Weaker regularity on f . If $f \in C^{0,\alpha}(\Gamma)$ (Hölder), the limits hold pointwise for all $\zeta_0 \in \Gamma$. If $f \in L^p(\Gamma)$, $1 < p < \infty$, the jump relations hold in the L^p (and a.e.) sense.

Example (unit circle). For $\Gamma = \{|\zeta| = 1\}$ and $f(\zeta) = \zeta^m$ ($m \geq 0$):

$$F(z) = \begin{cases} z^m, & |z| < 1, \\ 0, & |z| > 1, \end{cases} \quad \Rightarrow \quad F^+(e^{it}) - F^-(e^{it}) = e^{imt} = f(e^{it}),$$

and P. V. $\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - e^{it}} d\zeta = \frac{1}{2} e^{imt}$.

(ii) Curve only piecewise smooth / Lipschitz (corners allowed). The Plemelj formulas remain valid for *Lipschitz* curves (e.g. piecewise C^1 with finite length) when boundary limits are taken *nontangentially*. This is classical in singular–integral theory (Calderón–Zygmund): the Cauchy singular integral operator is bounded on $L^p(\Gamma)$ for $1 < p < \infty$, and the nontangential boundary limits exist almost everywhere.

(iii) Regularity of f .

- For classical *pointwise* limits at every boundary point, it suffices to assume $f \in C^{0,\alpha}(\Gamma)$ (Hölder) for some $\alpha \in (0, 1]$.
- For *almost-everywhere* (a.e.) limits and L^p jump relations, it is enough to assume $f \in L^p(\Gamma)$ with $1 < p < \infty$.
- The original hypothesis $f \in C^1(\Gamma)$ is more than enough; it guarantees the principal value is well defined and that the interior/exterior limits exist *at every* $\zeta_0 \in \Gamma$ (no exceptional set).

Cauchy integrals on the unit circle: analysis of the integrands and evaluation

Problem

Let γ be the positively oriented unit circle $\{|z| = 1\}$. Use Cauchy Integral Formulas to evaluate

$$(a) \quad \int_{\gamma} \frac{e^z \cos z}{z} dz, \quad (b) \quad \int_{\gamma} \frac{e^z \cos z}{z^3} dz.$$

Background

- **Cauchy Integral Formula (CIF).** If f is holomorphic on and inside γ and a lies inside, then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - a} d\zeta = f(a), \quad \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta = \frac{f^{(n)}(a)}{n!}.$$

- **Residues.** $\int_{\gamma} g(z) dz = 2\pi i \sum \text{Res}(g; \text{poles inside } \gamma).$
- e^z and $\cos z$ are entire; hence $N(z) := e^z \cos z$ is entire.

Anatomy of the integrands

Let $N(z) = e^z \cos z$.

- **Zeros of N .** e^z has no zeros; $\cos z$ has zeros at $z = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$ (simple zeros), none on $|z| = 1$.
- **Poles.** $\frac{N(z)}{z}$ has a simple pole at $z = 0$. $\frac{N(z)}{z^3}$ has a triple pole at $z = 0$. Because $N(0) = 1 \neq 0$, the pole orders are not reduced.
- **Domains.** Both $\frac{N}{z}$ and $\frac{N}{z^3}$ are holomorphic on $\mathbb{C} \setminus \{0\}$; the only singularity inside γ is at 0.
- **Magnitude along lines.** With $z = x + iy$,

$$|N(z)| = e^x |\cos(x + iy)|, \quad |\cos(x + iy)|^2 = \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y.$$

Thus along vertical lines (x fixed) $|N| \sim e^x \frac{e^{|y|}}{2}$ for large $|y|$; along horizontal lines (y fixed) $|N|$ oscillates with an e^x envelope.

Evaluations

(a) **The integral of $\frac{N(z)}{z}$.** By CIF with $f = N$ and $a = 0$,

$$\int_{\gamma} \frac{e^z \cos z}{z} dz = 2\pi i N(0) = 2\pi i e^0 \cos 0 = 2\pi i.$$

(b) **The integral of $\frac{N(z)}{z^3}$.** By the higher-order CIF with $n = 2$,

$$\int_{\gamma} \frac{e^z \cos z}{z^3} dz = 2\pi i \frac{N^{(2)}(0)}{2!}.$$

Compute derivatives:

$$N'(z) = e^z(\cos z - \sin z), \quad N''(z) = e^z(\cos z - \sin z) + e^z(-\sin z - \cos z) = -2e^z \sin z,$$

so $N''(0) = 0$. Therefore

$$\int_{\gamma} \frac{e^z \cos z}{z^3} dz = 0.$$

Remarks

- Laurent expansions at 0 show directly that the coefficient of $1/z$ in $\frac{N}{z^3}$ is 0; hence the integral vanishes.
- The jump/limit issues do not arise here since the integrands are holomorphic on and inside γ except at 0, which is isolated and handled by CIF.

Detailed anatomy of $N(z) = e^z \cos z$

Algebraic forms.

$$N(z) = e^z \cos z = \frac{1}{2} \left(e^{(1+i)z} + e^{(1-i)z} \right).$$

Thus N is entire of order 1 and exponential type $\sqrt{2}$.

Zeros and critical points.

- Zeros: $e^z \neq 0$, so $N(z) = 0 \iff \cos z = 0 \iff z = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$ (all simple).
- Critical points: $N'(z) = e^z(\cos z - \sin z) = 0 \iff \tan z = 1 \iff z = \frac{\pi}{4} + k\pi$ (nondegenerate since $N''(z) = -2e^z \sin z \neq 0$ there).
- Singularities: none (entire).

Real/imag parts and modulus. For $z = x + iy$,

$$e^z = e^x(\cos y + i \sin y), \quad \cos(x + iy) = \cos x \cosh y - i \sin x \sinh y,$$

so

$$\begin{aligned} N(z) = e^x & \left(\cos y \cos x \cosh y + \sin y \sin x \sinh y \right) \\ & + i e^x \left(\sin y \cos x \cosh y - \cos y \sin x \sinh y \right), \end{aligned}$$

and

$$|N(x + iy)|^2 = e^{2x} \left(\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y \right).$$

Vertical lines (x fixed). As $|y| \rightarrow \infty$, $\cosh y \sim \sinh y \sim \frac{1}{2}e^{|y|}$, hence

$$|N(x + iy)| \sim \frac{1}{2} e^{x+|y|} \quad (|y| \rightarrow \infty).$$

Horizontal lines (y fixed). Using $|\cos(x + iy)| \in [|\sinh y|, |\cosh y|]$ as x varies,

$$|N(x + iy)| \in [e^x |\sinh y|, e^x |\cosh y|],$$

so $|N(x + iy)|$ oscillates in x with an e^x envelope (growth for $x \rightarrow +\infty$, decay for $x \rightarrow -\infty$).

Directional growth (indicator). Along $z = re^{i\theta}$,

$$\log |N(re^{i\theta})| \approx r \max\{\cos \theta + \sin \theta, \cos \theta - \sin \theta\},$$

so the indicator is $h(\theta) = \max\{\cos \theta + \sin \theta, \cos \theta - \sin \theta\}$, with maximum $\sqrt{2}$ at $\theta = \pm \frac{\pi}{4}$.

Series at 0.

$$e^z \cos z = 1 + z + \underbrace{0}_{z^2} + \frac{z^3}{3} + O(z^4),$$

implying, for example,

$$\int_{|z|=1} \frac{e^z \cos z}{z^3} dz = 2\pi i [z^{-1}] \left(\frac{e^z \cos z}{z^3} \right) = 0.$$

Exponential integrands on the unit circle

Anatomy of $E(z) = e^z$

- **Entire/type.** E is entire of order 1 and exponential type 1.
- **Zeros/critical points.** No zeros; $E'(z) = e^z \neq 0$ everywhere, so no critical points.
- **Modulus and lines.** For $z = x + iy$, $|E(z)| = e^x$. Along vertical lines (x fixed) the modulus is constant and the argument increases linearly with y . Along horizontal lines (y fixed) the modulus is e^x (exponential growth/decay).
- **Directional growth.** On the ray $z = re^{i\theta}$, $|E(re^{i\theta})| = e^{r \cos \theta}$; indicator $h(\theta) = \cos \theta$.
- **Series at 0.** $e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \dots$.

Cauchy evaluations on $\gamma = \{|z| = 1\}$

Let $f(z) = e^z$.

(a) Using the Cauchy Integral Formula with $a = 0$,

$$\int_{\gamma} \frac{e^z}{z} dz = 2\pi i f(0) = 2\pi i.$$

(b) Using the higher-order Cauchy formula with $n = 2$,

$$\int_{\gamma} \frac{e^z}{z^3} dz = 2\pi i \frac{f^{(2)}(0)}{2!} = 2\pi i \cdot \frac{1}{2} = \pi i.$$

(Here $f^{(2)}(0) = e^0 = 1$. Equivalently, the z^{-1} -coefficient in the Laurent expansion of e^z/z^3 is $1/2$.)

Problem

Let γ be the positively oriented unit circle $\{|z| = 1\}$. Use the Cauchy Integral Formula to evaluate

$$(a) \quad \int_{\gamma} \frac{e^z \cos z}{z} dz, \quad (b) \quad \int_{\gamma} \frac{e^z \cos z}{z^3} dz.$$

Background and method

We recall two standard versions of the Cauchy Integral Formula.

Cauchy Integral Formula (CIF). If f is holomorphic in a neighborhood of the closed contour γ and a is a point inside γ , then

$$f(a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{\zeta - a} d\zeta.$$

Equivalently,

$$\oint_{\gamma} \frac{f(\zeta)}{\zeta - a} d\zeta = 2\pi i f(a).$$

Higher-order Cauchy formula. If f is holomorphic near a , then for $n \geq 0$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta,$$

or equivalently,

$$\oint_{\gamma} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta = 2\pi i \frac{f^{(n)}(a)}{n!}.$$

—

Structure of the integrand

Let

$$f(z) = e^z \cos z.$$

Then f is *entire*, since both e^z and $\cos z$ are entire.

Zeros. $e^z \neq 0$ everywhere, and $\cos z = 0$ at $z = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$. None of these zeros lie on $|z| = 1$, so f has no singularities on γ .

Singularity structure of the integrands.

$\frac{f(z)}{z}$ has a simple pole at $z = 0$, $\frac{f(z)}{z^3}$ has a triple pole at $z = 0$.

Since $f(0) = e^0 \cos 0 = 1 \neq 0$, the pole orders are not reduced.

—

(a) Evaluation of $\int_{\gamma} \frac{e^z \cos z}{z} dz$

By the Cauchy Integral Formula with $a = 0$, we have

$$\oint_{\gamma} \frac{f(z)}{z} dz = 2\pi i f(0).$$

Compute $f(0)$:

$$f(0) = e^0 \cos 0 = 1.$$

Therefore,

$$\boxed{\int_{\gamma} \frac{e^z \cos z}{z} dz = 2\pi i.}$$

—

(b) Evaluation of $\int_{\gamma} \frac{e^z \cos z}{z^3} dz$

By the higher-order Cauchy formula with $a = 0$ and $n = 2$,

$$\oint_{\gamma} \frac{f(z)}{z^3} dz = 2\pi i \frac{f^{(2)}(0)}{2!}.$$

Compute the derivatives.

$$f'(z) = e^z(\cos z - \sin z), \quad f''(z) = e^z(\cos z - \sin z) + e^z(-\sin z - \cos z) = -2e^z \sin z.$$

Thus,

$$f''(0) = -2e^0 \sin 0 = 0.$$

Hence

$$\boxed{\int_{\gamma} \frac{e^z \cos z}{z^3} dz = 0.}$$

—

Alternative check using the series expansion

We can verify the result directly from the Maclaurin series.

$$e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \cdots, \quad \cos z = 1 - \frac{z^2}{2} + \frac{z^4}{24} - \cdots.$$

Multiplying,

$$e^z \cos z = (1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \cdots)(1 - \frac{z^2}{2} + \frac{z^4}{24} - \cdots) = 1 + z + 0 \cdot z^2 + \frac{1}{3}z^3 + O(z^4).$$

Then

$$\frac{e^z \cos z}{z^3} = \frac{1}{z^3} + \frac{1}{z^2} + 0 \cdot \frac{1}{z} + \frac{1}{3} + O(z),$$

and the coefficient of $\frac{1}{z}$ is zero. By Cauchy's theorem (or the residue theorem),

$$\oint_{\gamma} \frac{e^z \cos z}{z^3} dz = 2\pi i [z^{-1}] \left(\frac{e^z \cos z}{z^3} \right) = 0.$$

—

Summary and interpretation

$(a) \quad \int_{\gamma} \frac{e^z \cos z}{z} dz = 2\pi i,$ $(b) \quad \int_{\gamma} \frac{e^z \cos z}{z^3} dz = 0.$
--

Geometric intuition. - The integrands have their only singularity at the origin, entirely enclosed by the contour γ . - The first integral measures the value of $f(0)$ via the Cauchy kernel $1/z$. - The second integral involves the second derivative of f at 0; since $f''(0) = 0$, its contribution vanishes. - In the Laurent expansion viewpoint, the integral picks out the coefficient of $\frac{1}{z}$ in $\frac{f(z)}{z^3}$, which here is zero.

Mean value property and maximum modulus principle

Problem (exact)

Let $f : U \rightarrow \mathbb{C}$ be holomorphic on a domain U .

(a) Show that, if $\overline{B}(a, r) \subset U$, then

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt.$$

- (b) Suppose $a \in U$ is such that $|f(a)| = \max_U |f|$. Show that $|f|$ must be constant near a .
- (c) Deduce that f is constant on all of U . *Hint:* Consider $S = \{z \in U : |f(z)| = |f(a)|\}$.

Background

Cauchy integral formula on circles. If $\overline{B}(a, r) \subset U$, then

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta-a|=r} \frac{f(\zeta)}{\zeta-a} d\zeta.$$

With $\zeta = a + re^{it}$ and $d\zeta = ire^{it} dt$, this becomes the circular average.

Equality in the triangle inequality. For an integrable complex function g , $|\int g| \leq \int |g|$ with equality iff g has a constant argument almost everywhere on the path of integration.

Cauchy integral for derivatives. If f is holomorphic on $\overline{B}(a, r)$, then for $n \geq 1$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{|\zeta-a|=r} \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta.$$

Solution

(a) Mean value property. By Cauchy's formula,

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta-a|=r} \frac{f(\zeta)}{\zeta-a} d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{it})}{re^{it}} (ire^{it}) dt = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt.$$

(b) Local constancy at an interior maximum. Assume $|f(a)| = \max_U |f|$. For $r > 0$ with $\overline{B}(a, r) \subset U$,

$$|f(a)| = \left| \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt \right| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{it})| dt \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a)| dt = |f(a)|.$$

Thus equality holds throughout. Hence $|f(a + re^{it})| = |f(a)|$ and $f(a + re^{it})$ has a constant argument (the same as $f(a)$) for almost every t . Therefore

$$f(a + re^{it}) = f(a) \quad \text{for all } t,$$

i.e. f is constant on the circle $|z - a| = r$. For such an r , Cauchy's formula for $n \geq 1$ gives

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{|\zeta-a|=r} \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta = \frac{n!}{2\pi i} f(a) \int_{|\zeta-a|=r} \frac{1}{(\zeta-a)^{n+1}} d\zeta = 0.$$

Hence the Taylor series of f at a is the constant $f(a)$; there exists $\rho > 0$ such that $f \equiv f(a)$ on $B(a, \rho)$. In particular, $|f|$ is constant near a .

(c) Constancy on the whole domain. Let $M = |f(a)|$ and $S = \{z \in U : |f(z)| = M\}$. The set S is closed (preimage of $\{M\}$ under the continuous map $z \mapsto |f(z)|$). By part (b), for each $z \in S$ there exists a neighborhood on which f is constant, hence that neighborhood lies in S ; thus S is open. Since U is connected and $S \neq \emptyset$, we conclude $S = U$, so $|f|$ is constant on U , and by (b) (applied at any point) f is constant on U .

Examples

- For $f(z) = z^n$, $f(0) = 0 = \frac{1}{2\pi} \int_0^{2\pi} (re^{it})^n dt$ (mean value).
- For $f(z) = e^z$ on any disk, $\max_{\overline{B}(a,r)} |f|$ occurs on $|z - a| = r$ (no interior maximum unless f is constant).

Mean Value Property for Holomorphic and Harmonic Functions

Statement (Holomorphic Case)

Let f be holomorphic on a domain $U \subset \mathbb{C}$. If the closed disk $\overline{B}(a, r) \subset U$, then

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt$$

i.e. the value of f at the center equals the average of its values on the surrounding circle.

One-Line Proof

By Cauchy's integral formula on the circle $|\zeta - a| = r$,

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{\zeta - a} d\zeta.$$

Parametrize $\zeta = a + re^{it}$ so that $d\zeta = ire^{it} dt$. Then

$$f(a) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{it})}{re^{it}} (ire^{it}) dt = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt.$$

Harmonic Version

If u is harmonic on U and $\overline{B}(a, r) \subset U$, then

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it}) dt$$

(For $f = u + iv$ holomorphic, both u and v are harmonic and satisfy this property.)

Intuition and Consequences

Rigidity / No interior spikes. The center value is a circular average; thus a modulus maximum cannot occur inside unless f is constant (maximum modulus principle).

Cauchy estimates. From circular averages and Cauchy formulas for derivatives, one gets the standard bounds

$$|f^{(n)}(a)| \leq \frac{n!}{r^n} \max_{|\zeta-a|=r} |f(\zeta)|.$$

Uniqueness. For harmonic functions, boundary data determines the interior via Poisson's formula; the mean value property is a key equivalent.

Examples

1) Polynomials. For $f(z) = z^n$ and $a = 0$,

$$\frac{1}{2\pi} \int_0^{2\pi} (re^{it})^n dt = \begin{cases} 1, & n = 0, \\ 0, & n \geq 1, \end{cases} = f(0).$$

2) Exponential. Let $f(z) = e^z$. Then on $|z - a| = r$,

$$f(a + re^{it}) = e^a e^{re^{it}} = e^a \sum_{k=0}^{\infty} \frac{r^k e^{ikt}}{k!}.$$

Averaging over $t \in [0, 2\pi]$ kills all $k \geq 1$, leaving

$$\frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt = e^a = f(a).$$

3) Harmonic example. If $u(x, y) = x$ (harmonic), then for any circle centered at $a = x_0 + iy_0$,

$$\frac{1}{2\pi} \int_0^{2\pi} u(a + re^{it}) dt = \frac{1}{2\pi} \int_0^{2\pi} (x_0 + r \cos t) dt = x_0 = u(a).$$

Equality Case and Maximum Modulus

If $|f(a)| = \max_{\overline{B}(a,r)} |f|$, then

$$|f(a)| = \left| \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt \right| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{it})| dt \leq |f(a)|.$$

Equality forces $f(a + re^{it})$ to have *constant argument* and constant modulus on the circle, hence f is constant on that circle. Cauchy's formula for derivatives then yields $f^{(n)}(a) = 0$ for $n \geq 1$, so f is constant near a ; on a connected domain, f is constant everywhere.

Harmonic vs. Holomorphic: What Implies What?

Holomorphic $\Rightarrow$ Harmonic

Let $f = u + iv$ be holomorphic on $U \subset \mathbb{C}$. Then $u, v \in C^\infty(U)$ and satisfy the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Differentiating and adding,

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0,$$

so u and v are *harmonic*. Hence every holomorphic function has harmonic real and imaginary parts.

Harmonic $\nRightarrow$ Holomorphic (in general)

A real-valued harmonic function u is not itself holomorphic (nonconstant real-valued holomorphic maps do not exist by the Open Mapping Theorem).

For a complex-valued $f = u + iv$ to be holomorphic one needs both:

$$(i) \Delta u = \Delta v = 0 \quad \text{and} \quad (ii) u_x = v_y, \quad u_y = -v_x.$$

Harmonicity alone does not enforce the Cauchy–Riemann relations.

Harmonic Conjugates and Topology

Given harmonic u on U , define the 1-form

$$\alpha := -u_y dx + u_x dy.$$

Then $d\alpha = (u_{xx} + u_{yy}) dx \wedge dy = \Delta u dx \wedge dy = 0$, so α is *closed*.

Local existence. Closed $\Rightarrow$ exact locally: there exists v (locally) with $dv = \alpha$, i.e.

$$v_y = u_x, \quad v_x = -u_y.$$

Thus $f = u + iv$ is *locally* holomorphic.

Global existence. If U is *simply connected*, closed $\Rightarrow$ exact globally, so there is a global harmonic conjugate v and $f = u + iv$ is globally holomorphic (unique up to an additive constant in v). If U is not simply connected, a global v exists iff all periods of α vanish:

$$\oint_{\gamma} \alpha = 0 \quad \text{for every closed loop } \gamma \subset U.$$

Examples

Holomorphic $\Rightarrow$ harmonic. $f(z) = e^z$; then $u = e^x \cos y$ and $v = e^x \sin y$ satisfy $\Delta u = \Delta v = 0$.

Harmonic but no global holomorphic primitive on $\mathbb{C} \setminus \{0\}$. $u(z) = \log |z|$ is harmonic on $\mathbb{C} \setminus \{0\}$. Locally $u = \Re(\log z)$, but a global conjugate would be $\arg z$, which is multivalued; the periods are nonzero, so no global single-valued holomorphic F with $\Re F = u$.

Harmonic with a global conjugate. $u(x, y) = x^2 - y^2$ is harmonic; $v(x, y) = 2xy$ satisfies the CR relations, and $f = u + iv = (x + iy)^2 = z^2$ is holomorphic.

Harmonic components but not holomorphic. $f(z) = \bar{z} = x - iy$. Here $u = x$, $v = -y$ are harmonic, but $u_x = 1 \neq v_y = -1$; the CR equations fail, so f is not holomorphic (it is anti-holomorphic).

Bottom Line

- **Always:** holomorphic $\Rightarrow$ (real and imaginary parts) harmonic.
- **Locally:** harmonic u is the real part of a holomorphic function.
- **Globally:** harmonic u is the real part of a holomorphic function *iff* the domain is simply connected (or, equivalently, the periods of $-u_y dx + u_x dy$ vanish).
- **Not sufficient:** harmonicity alone does not imply holomorphicity; the Cauchy–Riemann equations are the extra condition.

Product-of-distances lower bound on the unit circle

Problem

Let $w_1, \dots, w_n$ lie on the unit circle $\{|z| = 1\}$. Show that there exists z on the unit circle such that

$$\prod_{k=1}^n |z - w_k| \geq 1.$$

Key observation: average of $\log |z - w|$ is zero on the circle

Fix w with $|w| = 1$ and parametrize the circle by $z = e^{it}$, $t \in [0, 2\pi]$. Writing $w = e^{i\phi}$,

$$|e^{it} - w| = |e^{it} - e^{i\phi}| = 2 \left| \sin \frac{t - \phi}{2} \right|.$$

Hence

$$\frac{1}{2\pi} \int_0^{2\pi} \log |e^{it} - w| dt = \frac{1}{2\pi} \int_0^{2\pi} \log \left(2 \left| \sin \frac{t - \phi}{2} \right| \right) dt = \frac{1}{2\pi} \int_0^{2\pi} \log \left(2 \left| \sin \frac{u}{2} \right| \right) du = 0,$$

where the shift $u = t - \phi$ and the classical integral $\int_0^{2\pi} \log \left(2 \sin \frac{u}{2} \right) du = 0$ were used.

Proof of the claim

Define the polynomial

$$P(z) = \prod_{k=1}^n (z - w_k).$$

For $z = e^{it}$ on the unit circle,

$$\log |P(e^{it})| = \sum_{k=1}^n \log |e^{it} - w_k|.$$

Averaging over $t \in [0, 2\pi]$ and using the previous calculation term-by-term,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |P(e^{it})| dt = \sum_{k=1}^n \frac{1}{2\pi} \int_0^{2\pi} \log |e^{it} - w_k| dt = 0.$$

If $\log |P(e^{it})| < 0$ for all t , the average would be negative. Therefore there exists t_0 with

$$\log |P(e^{it_0})| \geq 0 \iff |P(e^{it_0})| = \prod_{k=1}^n |e^{it_0} - w_k| \geq 1.$$

Setting $z = e^{it_0}$ completes the proof.

Remarks and variants

- Equivalently, by Jensen's formula applied to P (whose zeros lie on $|z| = 1$), the mean of $\log |P(e^{it})|$ over the unit circle equals $\log |P(0)| = 0$, since $|P(0)| = \prod |w_k| = 1$.
- For $n = 1$, choosing $z = -w_1$ gives $|z - w_1| = 2 \geq 1$. For general n , the statement guarantees at least one z with product ≥ 1 , though the maximizing z depends on the configuration of the w_k .

Entire function with nonnegative real part is constant

Problem

Let f be holomorphic on $\mathbb{C}$ and suppose $\operatorname{Re} f(z) \geq 0$ for all $z \in \mathbb{C}$. Show that f is constant. (*Hint: consider $\exp(-f(z))$.*)

Background

- **Liouville's Theorem.** Every bounded entire function is constant.
- If $\operatorname{Re} f \geq 0$ and $g(z) := e^{-f(z)}$, then

$$|g(z)| = |e^{-f(z)}| = e^{-\operatorname{Re} f(z)} \leq 1,$$

so g is entire and bounded.

Solution

Define $g(z) := e^{-f(z)}$. Then g is entire and satisfies $|g(z)| \leq 1$ for all $z \in \mathbb{C}$. By Liouville's theorem, g is constant: $g(z) \equiv c$ with $|c| \leq 1$.

Hence $e^{-f(z)} \equiv c$. Choosing any logarithm of the constant c ,

$$-f(z) \equiv \log c \quad \implies \quad f(z) \equiv -\log c,$$

so f is constant on $\mathbb{C}$.

Remark (optional)

If f were nonconstant, then $g = e^{-f}$ would be nonconstant and thus an open map. But $g(\mathbb{C}) \subset \{w : |w| \leq 1\}$ is not open, a contradiction. Therefore f must be constant.

Reverse assumption: $\operatorname{Re} f \leq 0$ implies f is constant

Claim

If f is entire and $\operatorname{Re} f(z) \leq 0$ for all $z \in \mathbb{C}$, then f is constant.

Proof

Define $h(z) := e^{f(z)}$. Then h is entire and

$$|h(z)| = |e^{f(z)}| = e^{\operatorname{Re} f(z)} \leq 1 \quad (\forall z \in \mathbb{C}).$$

Hence h is a bounded entire function. By Liouville's theorem, h is constant: $h(z) \equiv c$ with $|c| \leq 1$. Taking any logarithm of the constant c , we get $f(z) \equiv \log c$, a constant function.

Remark (generalization)

If $\operatorname{Re} f$ is bounded *above* or *below* on $\mathbb{C}$, then f is constant. Indeed, use e^f when $\operatorname{Re} f \leq M$ or e^{-f} when $\operatorname{Re} f \geq m$ to obtain a bounded entire function and apply Liouville.

Half-plane criteria for constancy of entire functions

General Lemma (rotation-shift half-plane test)

Let f be entire. Suppose there exist $\theta \in \mathbb{R}$ and $c \in \mathbb{R}$ such that

$$\Re(e^{-i\theta} f(z)) \geq c \quad (\forall z \in \mathbb{C}).$$

Then f is constant.

Proof. Define

$$g(z) := \exp(-e^{-i\theta}(f(z) - c)).$$

Then

$$|g(z)| = \exp\left(-\Re(e^{-i\theta}(f(z) - c))\right) \leq 1,$$

so g is bounded and entire. By Liouville's theorem, g is constant; hence f is constant. $\square$

Immediate corollaries (choose θ appropriately)

- **Real part bounded below.** If $\Re f(z) \geq a$ for all z , take $\theta = 0$:

$$|e^{-(f-a)}| = e^{-(\Re f - a)} \leq 1 \Rightarrow e^{-(f-a)} \text{ constant} \Rightarrow f \text{ constant.}$$

- **Real part bounded above.** If $\Re f(z) \leq a$ for all z , use e^{f-a} :

$$|e^{f-a}| = e^{\Re f - a} \leq 1 \Rightarrow e^{f-a} \text{ constant} \Rightarrow f \text{ constant.}$$

- **Imaginary part bounded below.** If $\Im f(z) \geq b$ for all z , take $\theta = \frac{\pi}{2}$:

$$|e^{if}| = e^{-\Im f} \leq e^{-b} \Rightarrow e^{if} \text{ constant} \Rightarrow f \text{ constant.}$$

- **Imaginary part bounded above.** If $\Im f(z) \leq b$ for all z , take $\theta = -\frac{\pi}{2}$:

$$|e^{-if}| = e^{\Im f} \leq e^b \Rightarrow e^{-if} \text{ constant} \Rightarrow f \text{ constant.}$$

- **Tilted half-planes.** If $\Re(e^{-i\theta} f(z)) \geq a$ (or $\leq a$) for some fixed θ , apply the lemma directly. Thus any entire map with image contained in a rotated/translated half-plane is constant.
-

Strip versions and harmonic viewpoint

If $a \leq \Re(e^{-i\theta} f) \leq b$ on $\mathbb{C}$, then $\Re(e^{-i\theta} f)$ is a bounded harmonic function on $\mathbb{C}$, hence constant. The Cauchy–Riemann equations then force the imaginary part to be constant as well, so f is constant.

Similarly, if $a \leq \Im(e^{-i\theta} f) \leq b$, apply the same argument to the harmonic function $\Im(e^{-i\theta} f)$.

Remarks

- The choices $e^{\pm f}$ and $e^{\pm if}$ are tailored so that their moduli are $e^{\pm \Re f}$ and $e^{\mp \Im f}$, respectively.
- This gives a unified proof that an entire function with image in *any* (open or closed) half-plane must be constant.
- Mapping a half-plane to the unit disk with a Möbius map (e.g. Cayley transform) yields the same conclusion: the composition is bounded entire, hence constant.

Growth and domination for entire functions

Problem

Let f be an entire function.

- (i) Prove that f is a polynomial of degree at most k if and only if there exist real constants $M, R > 0$ and an integer k such that

$$|f(z)| \leq M |z|^k \quad \text{for } |z| > R.$$

- (ii) Let g be an entire function such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f(z) = c g(z)$ for all $z \in \mathbb{C}$.
-

Background

Cauchy estimates. If f is holomorphic on $|z| \leq \rho$, then

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}} dz, \quad |f^{(n)}(0)| \leq \frac{n!}{\rho^n} \max_{|z|=\rho} |f(z)|.$$

Liouville and removable singularities. A bounded entire function is constant. If h is holomorphic on a punctured neighborhood of a and bounded near a , then h extends holomorphically to a .

Solution to (i)

“Only if.” If $f(z) = a_k z^k + \cdots + a_0$ has degree $\leq k$, then for $|z| > 1$,

$$|f(z)| \leq (|a_k| + \cdots + |a_0|) |z|^k.$$

Taking $R \geq 1$ and $M = \sum_{j=0}^k |a_j|$ gives the claim.

“If.” Assume there exist $M, R > 0$ and integer k with $|f(z)| \leq M|z|^k$ for $|z| > R$. Fix $\rho > R$. By Cauchy’s estimate,

$$|f^{(n)}(0)| \leq \frac{n!}{\rho^n} \max_{|z|=\rho} |f(z)| \leq \frac{n!}{\rho^n} M \rho^k = M n! \rho^{k-n}.$$

If $n > k$, then letting $\rho \rightarrow \infty$ yields $f^{(n)}(0) = 0$. Hence all derivatives of order $> k$ vanish, so f is a polynomial of degree at most k .

Solution to (ii)

If $g \equiv 0$, then $|f| \leq |g|$ forces $f \equiv 0$, and $f = cg$ is trivial. Assume $g \not\equiv 0$.

Define $h(z) = \frac{f(z)}{g(z)}$ where $g(z) \neq 0$. We show h extends to an entire, bounded function.

Let a be a zero of g of order $m \geq 1$, so $g(z) = (z-a)^m g_m(z)$ with g_m holomorphic and $g_m(a) \neq 0$. From $|f| \leq |g|$ we have, for z near a ,

$$|f(z)| \leq C |z-a|^m$$

for some $C > 0$, hence $f(z) = (z-a)^m f_m(z)$ with f_m holomorphic (and bounded) near a . Then on the punctured neighborhood,

$$h(z) = \frac{f(z)}{g(z)} = \frac{f_m(z)}{g_m(z)}.$$

Since $g_m(a) \neq 0$ and f_m is bounded near a , h is bounded near a ; by the removable singularities theorem, h extends holomorphically to a . Doing this at all zeros of g , we obtain an entire extension of h . Moreover, on the set where $g \neq 0$ we have $|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$, therefore by continuity $|h| \leq 1$ everywhere.

Thus h is bounded and entire, so by Liouville, $h \equiv c$ for some $c \in \mathbb{C}$ (with $|c| \leq 1$). Consequently $f(z) = c g(z)$ for all $z \in \mathbb{C}$.

Examples

- If $f(z) = 3z^5 - 2z + 7$ then $|f(z)| \leq (3 + 2 + 7)|z|^5$ for $|z| > 1$; the growth condition holds with $k = 5$.
- If $|f(z)| \leq |\sin z|$ on $\mathbb{C}$, then by (ii) there is $c \in \mathbb{C}$ with $f(z) = c \sin z$ for all z .

Poisson and Schwarz Integral Formulas on the Unit Disk

Let $f = u + iv$ be holomorphic in a domain containing the closed unit disk $\overline{\mathbb{D}}$.

(a) Show that

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) dt.$$

(b) Assume that for every $z_0 \in \mathbb{D}$ the map

$$h_{z_0}(z) = \frac{z + z_0}{1 + \overline{z_0}z}$$

is a bijection of $\mathbb{D}$ onto itself. Using this map, show that for $z \in \mathbb{D}$,

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \frac{1 - |z|^2}{|e^{it} - z|^2} dt.$$

(c) Using the fact that

$$\frac{|w|^2 - |z|^2}{|z - w|^2} = \Re\left(\frac{w + z}{w - z}\right),$$

show that

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \frac{e^{it} + z}{e^{it} - z} dt + i v(0).$$

Background (first theory)

Mean value property. If u is harmonic on a neighborhood of $\overline{\mathbb{D}}$, then for $0 < r \leq 1$,

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(re^{it}) dt.$$

Poisson kernel on $\mathbb{D}$. For $z = re^{i\phi} \in \mathbb{D}$ and boundary point e^{it} ,

$$P_r(\phi - t) = \frac{1 - r^2}{|e^{it} - re^{i\phi}|^2} = \frac{1 - |z|^2}{|e^{it} - z|^2}.$$

If $g \in C(\partial\mathbb{D})$ and U is harmonic in $\mathbb{D}$ with nontangential boundary values g , then

$$U(z) = \frac{1}{2\pi} \int_0^{2\pi} g(e^{it}) P_r(\phi - t) dt.$$

Disk automorphisms. For $z_0 \in \mathbb{D}$,

$$h_{z_0}(z) = \frac{z + z_0}{1 + \overline{z_0}z}, \quad h_{z_0}(0) = z_0, \quad h_{z_0}^{-1} = h_{-z_0},$$

is a Möbius automorphism of $\mathbb{D}$ carrying $\partial\mathbb{D}$ onto itself and preserving harmonic measure.

Schwarz/Herglotz kernel. For $z \in \mathbb{D}$,

$$K(z, e^{it}) = \frac{e^{it} + z}{e^{it} - z}, \quad \Re K(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2} = P_r(\phi - t).$$

Solutions

(a) Mean value at the center

Since f is holomorphic near $\overline{\mathbb{D}}$, its real part u is harmonic there. The mean value property at the origin gives

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) dt.$$

(b) Poisson integral via h_{z_0}

Fix $z \in \mathbb{D}$ and put $z_0 = z$. Consider the harmonic function

$$U(\zeta) := u(h_z(\zeta)), \quad \zeta \in \mathbb{D}.$$

By (a), applied at the origin for U ,

$$U(0) = \frac{1}{2\pi} \int_0^{2\pi} U(e^{it}) dt = \frac{1}{2\pi} \int_0^{2\pi} u(h_z(e^{it})) dt.$$

But $U(0) = u(h_z(0)) = u(z)$. Using the boundary change of variables $e^{i\theta} = h_z(e^{it})$ and the known identity

$$\frac{dt}{d\theta} = \frac{1 - |z|^2}{|e^{i\theta} - z|^2},$$

we obtain

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\theta}) \frac{1 - |z|^2}{|e^{i\theta} - z|^2} d\theta,$$

which is the Poisson integral formula.

(c) Schwarz integral for f

Define

$$F(z) := \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \frac{e^{it} + z}{e^{it} - z} dt.$$

For fixed $z \in \mathbb{D}$ the integrand is holomorphic in z and integrable in t , so F is holomorphic on $\mathbb{D}$. Taking real parts and using

$$\Re\left(\frac{e^{it} + z}{e^{it} - z}\right) = \frac{1 - |z|^2}{|e^{it} - z|^2},$$

we get

$$\Re F(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \frac{1 - |z|^2}{|e^{it} - z|^2} dt = u(z)$$

by part (b). Hence $F - f$ is holomorphic with $\Re(F - f) \equiv 0$ on $\mathbb{D}$, so $F - f \equiv iC$ for some real constant C . Evaluating at $z = 0$,

$$F(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) dt = u(0),$$

so

$$u(0) + iC = f(0) = u(0) + iv(0) \quad \Rightarrow \quad C = v(0).$$

Therefore

$$f(z) = F(z) + i v(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it}) \frac{e^{it} + z}{e^{it} - z} dt + i v(0).$$

Normalization and quick intuition

Normalize once:

$$K_t(z) = C(e^{-it}z), \quad C(\zeta) = \frac{1 + \zeta}{1 - \zeta} \quad (\text{Cayley transform}).$$

So every geometric fact for K_t is a rotation of the corresponding fact for C (which “focuses” at the boundary point 1).

Natural domain and image

$$\boxed{\mathbb{D} = \{z : |z| < 1\}}, \quad \boxed{K_t(\mathbb{D}) = \{w : \Re w > 0\} \text{ (right half-plane)}}.$$

Reason: for $|z| < 1$,

$$\Re K_t(z) = \frac{1 - |z|^2}{|e^{it} - z|^2} > 0.$$

Boundary behavior.

$$K_t(0) = 1, \quad \lim_{z \rightarrow e^{it}} K_t(z) = \infty, \quad K_t(\partial\mathbb{D} \setminus \{e^{it}\}) = i\mathbb{R}.$$

Thus $K_t : \mathbb{D} \rightarrow \{\Re w > 0\}$ is a conformal bijection, sending the boundary point e^{it} to ∞ and preserving orientation.

Inverse and derivative (for checks)

$$K_t^{-1}(w) = e^{it} \frac{w-1}{w+1}, \quad K'_t(z) = \frac{2e^{it}}{(e^{it}-z)^2}, \quad |K'_t(z)| = \frac{2}{|e^{it}-z|^2}.$$

Vertical lines in the image and their preimages

Fix $c > 0$. The vertical line

$$L_c = \{ w : \operatorname{Re} w = c \}$$

pulls back to the level set

$$\operatorname{Re} K_t(z) = \frac{1-|z|^2}{|e^{it}-z|^2} = c.$$

Rotate by $u = e^{-it}z$ (so $e^{it} \mapsto 1$). For the Cayley map C this is

$$\operatorname{Re} C(u) = \frac{1-|u|^2}{|1-u|^2} = c \iff |1-u|^2 = c^{-1}(1-|u|^2).$$

Writing $u = x + iy$ gives the circle equation

$$\boxed{\left(x - \frac{1}{1+c}\right)^2 + y^2 = \left(\frac{1}{1+c}\right)^2}.$$

Hence in the u -plane we get a circle of radius $1/(1+c)$ centered at $1/(1+c)$ on the real axis, tangent to $\partial\mathbb{D}$ at $u = 1$. Rotating back:

$$\boxed{\{\operatorname{Re} K_t(z) = c\} \text{ is a circle in } \mathbb{D} \text{ tangent to } \partial\mathbb{D} \text{ at } \tau = e^{it}.$$

As $c \downarrow 0$ the circle expands to $\partial\mathbb{D}$ (except at τ); as $c \uparrow \infty$ it shrinks to τ .

Vertical bands. The strip $a < \operatorname{Re} w < b$ corresponds to the lens-like region in $\mathbb{D}$ between the two tangent circles $\operatorname{Re} K_t(z) = a$ and $\operatorname{Re} K_t(z) = b$ (both tangent at τ).

Horizontal lines in the image and their preimages

Fix $b \in \mathbb{R}$. The horizontal line

$$H_b = \{ w : \operatorname{Im} w = b \}$$

pulls back to $\operatorname{Im} K_t(z) = b$. For the Cayley transform,

$$\operatorname{Im} C(u) = \frac{2 \operatorname{Im} u}{|1-u|^2} = b \iff (1-\operatorname{Re} u)^2 + \left(\operatorname{Im} u - \frac{1}{b}\right)^2 = \left(\frac{1}{b}\right)^2.$$

Thus for $b \neq 0$,

$$\boxed{\{\operatorname{Im} C(u) = b\} \text{ is a circle tangent to } \partial\mathbb{D} \text{ at } u = 1, \text{ center } (1, 1/b), \text{ radius } 1/|b|.$$

Rotating back by e^{it} shows that $\{\operatorname{Im} K_t(z) = b\}$ is also a circle tangent to $\partial\mathbb{D}$ at $\tau = e^{it}$. The two families

$$\{\operatorname{Re} K_t = c\}_{c>0}, \quad \{\operatorname{Im} K_t = b\}_{b\in\mathbb{R}}$$

are orthogonal (conformality).

Horizontal bands. The strip $\beta < \operatorname{Im} w < \delta$ pulls back to the region between the circles $\operatorname{Im} K_t(z) = \beta$ and $\operatorname{Im} K_t(z) = \delta$ (both tangent at τ).

Canonical correspondences

- $z = 0 \mapsto w = 1$.
- $z = \tau = e^{it} \mapsto w = \infty$ (simple pole).
- $\partial\mathbb{D} \setminus \{\tau\} \mapsto i\mathbb{R}$.
- Vertical line $\operatorname{Re} w = c \iff$ circle $\operatorname{Re} K_t(z) = c$ tangent at τ .
- Horizontal line $\operatorname{Im} w = b \iff$ circle $\operatorname{Im} K_t(z) = b$ tangent at τ .
- Vertical/horizontal *bands* $\iff$ regions between pairs of tangent circles.

Harmonic meaning

$$\operatorname{Re} K_t(z) = \frac{1 - |z|^2}{|e^{it} - z|^2} = P(z, e^{it})$$

is exactly the Poisson kernel of the unit disk with pole at the boundary point e^{it} . Hence the circles $\operatorname{Re} K_t(z) = c$ are level sets of harmonic measure; $\operatorname{Im} K_t$ is (up to an additive constant in t) the harmonic conjugate, whose level sets form the orthogonal tangent pencil.

Setup

Let $\tau = e^{it}$ and note

$$K_t(z) = C(e^{-it}z), \quad C(\zeta) = \frac{1 + \zeta}{1 - \zeta}, \quad z = \frac{w - 1}{w + 1}.$$

“Vertical” and “horizontal” in the domain are taken *relative to* τ :

$$\text{vertical: } \Re(e^{-it}z) = a, \quad \text{horizontal: } \Im(e^{-it}z) = b.$$

Images of vertical domain lines and bands

Impose $\Re z = a$ with $z = (w - 1)/(w + 1)$. Writing $w = u + iv$,

$$\Re\left(\frac{w-1}{w+1}\right) = a \iff \frac{|w|^2 - 1}{|w+1|^2} = a \iff \left(u - \frac{a}{1-a}\right)^2 + v^2 = \left(\frac{1}{1-a}\right)^2.$$

Thus

$$\boxed{\Re(e^{-it}z) = a \implies K_t(z) \text{ lies on the circle } \left(u - \frac{a}{1-a}\right)^2 + v^2 = \left(\frac{1}{1-a}\right)^2.}$$

In particular:

- $a = 0$ maps to the unit circle $|w| = 1$.
- $a \rightarrow 1^-$ produces circles whose limits are vertical lines in the right half-plane.
- The band $a < \Re(e^{-it}z) < b$ maps to the region between the two circles with centers $\frac{a}{1-a}$ and $\frac{b}{1-b}$ on the real axis and radii $1/|1-a|$, $1/|1-b|$.

Images of horizontal domain lines and bands

Impose $\Im z = b$ with $z = (w - 1)/(w + 1)$. Then

$$\Im\left(\frac{w-1}{w+1}\right) = b \iff \frac{2\Im w}{|w+1|^2} = b \iff (u+1)^2 + \left(v - \frac{1}{b}\right)^2 = \left(\frac{1}{b}\right)^2 \quad (b \neq 0).$$

Therefore

$$\boxed{\Im(e^{-it}z) = b \implies K_t(z) \text{ lies on the circle } (u+1)^2 + \left(v - \frac{1}{b}\right)^2 = \left(\frac{1}{b}\right)^2.}$$

Special cases:

- $b = 0$ maps to the positive real axis $\{w > 0\}$.
- For $b \neq 0$ the circle is tangent to the real axis at $w = -1$, with center $(-1, 1/b)$ and radius $1/|b|$.
- The band $\beta < \Im(e^{-it}z) < \delta$ maps to the region between the two tangent circles corresponding to $b = \beta$ and $b = \delta$.

Orthogonality and Möbius geometry

The families

$$\{\Re(e^{-it}z) = a\}_{a \in \mathbb{R}} \quad \text{and} \quad \{\Im(e^{-it}z) = b\}_{b \in \mathbb{R}}$$

are orthogonal in the z -plane, and K_t (being holomorphic/Möbius where defined) preserves angles. Hence their images are two orthogonal pencils of circles/lines in the w -plane.

Band dictionary (domain $\leftrightarrow$ image)

Domain band	$\iff$	Image region
$a < \Re(e^{-it}z) < b$	$\iff$	between two circles centered on the real axis
$\beta < \Im(e^{-it}z) < \delta$	$\iff$	between two circles tangent at $w = -1$

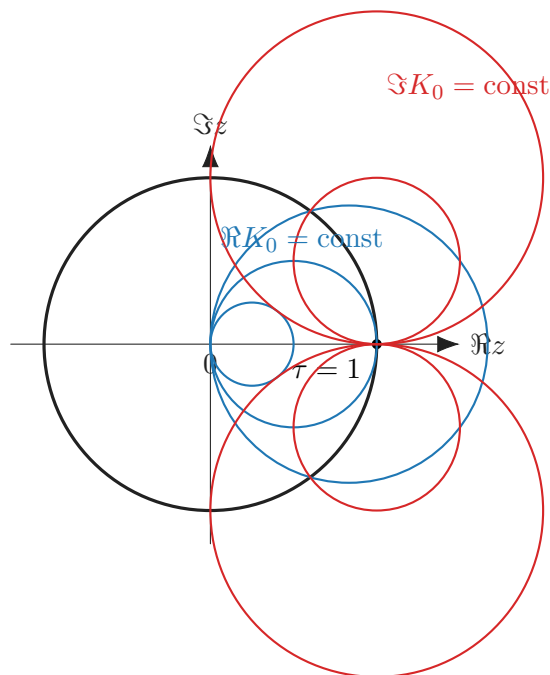


Figure 1: Level sets of $\Re K_0$ (blue) and $\Im K_0$ (red): tangent at 1, orthogonal elsewhere.

Poisson integral on the unit disk. If U is harmonic in $\mathbb{D}$ and has (nontangential) boundary values $g \in C(\partial\mathbb{D})$, then for $z = re^{i\phi} \in \mathbb{D}$,

$$U(re^{i\phi}) = \frac{1}{2\pi} \int_0^{2\pi} g(e^{it}) P_r(\phi - t) dt, \quad P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2} = \frac{1 - r^2}{|e^{i\theta} - r|^2}.$$

Here $P_r(\theta) \geq 0$ and $\frac{1}{2\pi} \int_0^{2\pi} P_r(\theta) d\theta = 1$, so the value at $re^{i\phi}$ is a weighted average of the boundary data on $\partial\mathbb{D}$. Moreover, $P_r(\phi - t) = \Re\left(\frac{e^{it} + re^{i\phi}}{e^{it} - re^{i\phi}}\right)$ and $U(0) = \frac{1}{2\pi} \int_0^{2\pi} g(e^{it}) dt$.

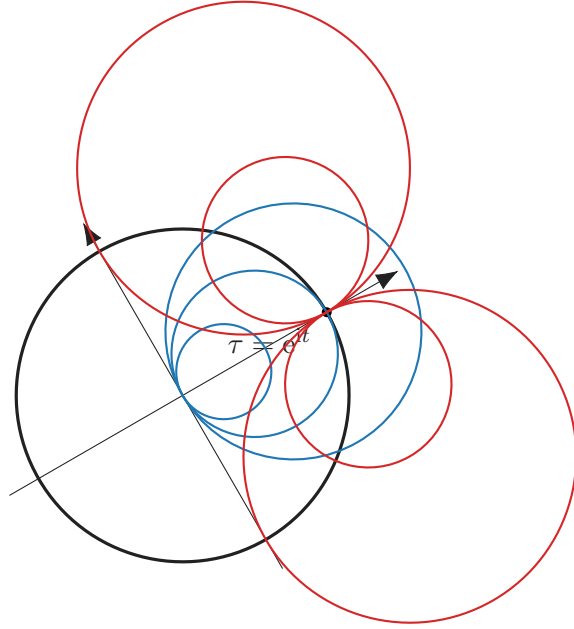


Figure 2: Same grids for K_t , obtained by rotating the $t = 0$ picture by angle t .

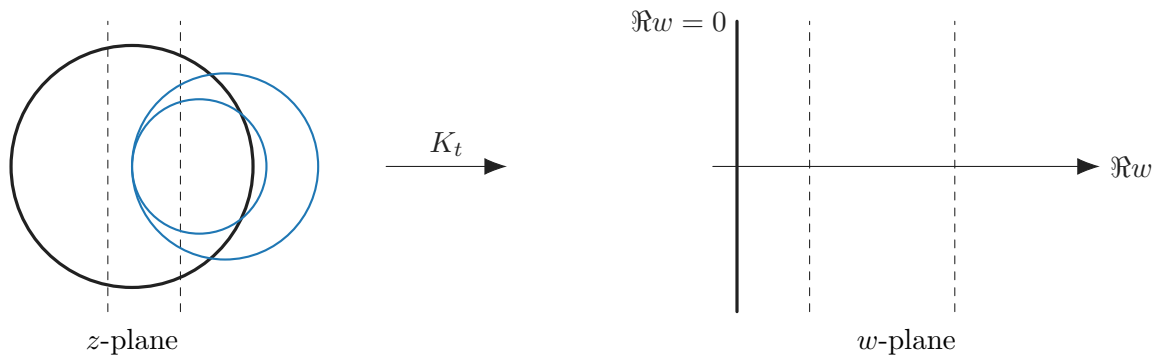


Figure 3: Domain band (between two vertical lines in z) $\mapsto$ vertical strip (between two lines) in the right half-plane.